

Data Centre Capacity Expansion to 2035

Data centre power demand is set to surge across Europe and the UK. Recent forecasts suggest dramatic growth: Oxford Economics projects UK data centre load rising from ~2.9 GW in 2024 to ~8.1 GW by 2030 [33†L238-L247] , while S&P Global forecasts European data centre demand doubling to ~36 GW by 2030 [45†L1538-L1546] . In the UK Parliament Library's assessment, Britain had about 1.6 GW of data centre capacity in 2024, with an additional 3.3–6.3 GW possible by 2030 under high-growth scenarios [9†L72-L80] [33†L238-L247] . Ireland's grid is even more concentrated: data centres already consumed 21–22% of Ireland's electricity in 2023–24 [11†L47-L51] [7†L111-L119] , and projections suggest this could reach ~30% by the early 2030s [12†L285-L293] [35†L112-L119] . Across the European Union, governments are actively courting hyperscale “AI” data centre builds: the EU's AI Action Plan aims to triple EU-wide data centre capacity within 5–7 years [45†L1549-L1558] . Much of this new capacity is aimed at AI computation, with individual “hyperscale” centres expected to draw as much power as 20,000–100,000 homes (hundreds of MW each) [17†L159-L167] [45†L1584-L1593] .

[73†embed_image] In the UK, planned data centre IT power capacity is forecast to more than double by 2030. For example, Oxford Economics projects UK data centres' IT load growing from about 2.9 GW (2024) to roughly 8.1 GW by 2030 [33†L238-L247] . These new facilities would raise UK data centre electricity use from ~2% of today's grid to nearly 9% by 2030 [33†L263-L266] . In Ireland, EirGrid and the energy regulator note a similar trend: data centres have risen from 5% of Ireland's demand in 2015 to ~21% in 2023 [11†L47-L51] [7†L111-L119] and could approach 30–33% by the mid-2030s [12†L285-L293] [7†L111-L119] . On the continent, data centre growth is being driven by EU policy and the search for “AI leadership.” S&P Global reports that European data centre demand was 18.7 GW in 2024 and is expected to climb to ~36 GW by 2030 [45†L1530-L1538] . This includes both on-premises and leased hyperscale centres, with Germany (4.26 GW), the UK (3.69 GW), and France (1.72 GW) already leading in 2025 [45†L1564-L1572] . Commissioned studies warn that meeting these jumps in power draw will require rapid grid expansion and policy changes [45†L1580-L1590] [14†L284-L293] .

Electricity Demand and Grid Impacts

Data centre expansion poses major challenges to electricity systems. In the UK, Ofgem warned that around 140 proposed data centre projects could collectively require ~50 GW of new power capacity – about 5 GW above the current national peak [37†L167-L174] . Such a surge risks straining the grid: Ofgem notes that connecting these loads could delay other projects (including renewables) and requires substantial investment in new substations and lines [37†L173-L181] [41†L203-L213] . In Ireland, EirGrid has repeatedly highlighted a tight supply-demand balance. Recent all-island adequacy studies project that Irish peak demand may exceed 7 GW by 2034 [46†L78-L85] [46†L90-L94] , driven largely by data centre growth (EirGrid forecasts that ~89% of demand growth 2024–34 is due to data centres [35†L112-L119]). With peak already topping 6 GW in January 2025 [46†L85-L94] , regulators required new Irish data centres to include on-site generation or storage just to connect [35†L112-L119] .

In continental Europe, grid bottlenecks are equally severe. Grid operators report multi-year waits (2–10 years) for data centre grid connections in many regions [14†L332-L340] . Some countries (e.g. Amsterdam) have even imposed moratoria on new high-load centres due to “a lack of power capacity” [14†L284-L293] [50†L79-L87] . The rapid rise in large loads forces alternative allocation

rules: Germany's federal grid regulator has moved from first-come-first-served to more restrictive allocation for city networks like Frankfurt and Berlin [52†L168-L176] . In practice, many new German centres are resorting to on-site gas generators to meet power needs, since the grid cannot supply the planned capacity [52†L190-L199] . For example, a Frankfurt data campus expanded from 84 MW to 126 MW by adding gas-fired generators alongside grid connections [52†L190-L197] . Experts warn that extensive on-site fossil generation (diesel or gas) for data centres will raise gas demand and CO₂ emissions, ultimately socializing costs (through higher gas prices and climate impacts) onto the community [52†L219-L228] .

The power needed for billions of kilowatt-hours per year of data centre load also has pricing implications. Utilities and analysts note two main cost drivers: the need for new generation (often gas-fired or renewable peakers) and the build-out of transmission/distribution infrastructure [41†L203-L213] . The costs of these investments typically feed into grid tariffs paid by all customers. In the US, modeling suggests data centre-driven infrastructure has been a key factor in electricity price rises: US prices are up ~42% since 2019, outpacing inflation [41†L219-L226] , while utilities have sought to pass tens of billions in grid upgrade costs onto ratepayers [39†L158-L166] . If similar dynamics occur in the UK and EU, households – especially poorer ones – will ultimately see higher bills for electricity, since even business rates relief or targeted subsidies cannot negate the underlying generation and network costs.

Water Use and Competition

Data centres are prodigious water users, with implications for water stress and affordability. Most data centres use evaporative cooling (cooling towers) that consume large volumes of fresh water. A representative estimate is that a 100 MW data centre can use ~2.5–2.9 billion litres of water per year [26†L281-L286] – roughly the annual needs of ~80,000 people. Indeed, IEA analysis notes global data centre water demand exceeds 0.5 trillion litres annually [26†L277-L284] , and could approach 1.2 trillion litres by 2030 if current trends continue. By one estimate, global AI data centre water use by 2027 (4.2–6.6 billion m³) would be about half of the UK's entire annual water consumption [26†L318-L326] . In absolute terms, large US centres have been measured at up to 5 million gallons (19 million litres) per day [17†L159-L167] .

This scale of freshwater demand can conflict with other users. In water-stressed regions, adding data centre demand intensifies competition between residential, agricultural, and industrial users [28†L631-L639] . For example, UK studies warn that current water resource plans (to 2050) scarcely account for AI and data centre demands [25†L17-L24] . By ignoring this demand, regulators risk exacerbating shortages: UK forecasts foresee a ~5 billion-litre daily deficit by 2050 (equivalent to over one-third of current supply) [25†L13-L18] , meaning even modest new industrial draws could force cutbacks. Officials and advocates have noted that data centres' water use often goes unpriced. In the UK, two-fifths of data centre operators do not even track their water withdrawals [23†L162-L167] . Without oversight, communities see clean water piped to private operators with little compensation. This has provoked public concern: one study notes that introducing large, uncounted data centre withdrawals in a finite basin is "likely to exacerbate" existing tensions between farmers, residents and industry [28†L631-L639] .

For low-income and rural communities, rising water scarcity and costs fall especially hard. Households typically spend a higher share of income on water if they are poorer [61†L445-L453] . For example, Ofwat's analysis shows the poorest third of UK households already spend about 3.7% of income on water/sewerage (vs a national average of ~1.6%) [61†L445-L453] . Any rate rises or restrictions driven by new industrial demand (like data centres) would thus hit these groups hardest. Rural areas are

particularly vulnerable: many rural residents rely on scarce groundwater or marginal supplies, and also disproportionately depend on agriculture, which competes for the same water. As one UK report warns, when water must be rationed in a drought, communities – including farmers – will clash over priority [25†L19-L24] . In drought-prone parts of the US, activists describe large tech “soda straw” projects sucking local aquifers dry [55†L290-L299] . By analogy, rural and low-income populations in Europe could face similar water cutoffs or price hikes if data centre cooling needs are not constrained.

Gas and Net-Zero Tension

Meeting data centre power demand often relies on gas-fired backup and generation, challenging decarbonisation targets. Even as renewables expand, grid operators foresee a need for flexible gas plants to meet sudden peaks and low-renewable intervals [46†L108-L116] . In Ireland’s forecasts, new “renewable gas-ready” generators are included to cover low-wind periods [46†L108-L116] . In practice, data centres themselves are increasingly turning to on-site gas. In Frankfurt, energy firms report DC projects installing gas turbine generators to add capacity [52†L190-L199] . Industry sources confirm this is not unique: many centres in Frankfurt and elsewhere see on-site gas plants as the only way to get power [52†L199-L208] . While companies frame this as a temporary or hydrogen-ready solution, experts caution it locks in fossil fuel use. Even German analysts note that if data centres rely on gas, consumers risk higher gas prices and greater import dependency [52†L219-L228] . Additional local impacts (noise, air pollution from turbines) are also shifted onto the community [52†L219-L228] .

The net effect is to partially reverse the clean-energy transition. UN and NGO warnings emphasize that without action, huge data centre growth could put climate goals at risk: UN Secretary-General Guterres warned that by 2030 data centres could consume as much power as all of Japan does today [45†L1591-L1599] , and called on firms to use 100% renewables and sustainable cooling [45†L1584-L1593] [45†L1591-L1599] . However, when grid capacity is insufficient, continuing expansion tends to be serviced by gas. In the UK, Ofgem warned that unbridled data centre growth could clash with the 2030 carbon-free target [37†L185-L193] . Europe’s planners now face a tension: allow massive new loads that may rely on new gas turbines, or constrain them to meet decarbonisation commitments.

Distributional Impacts and Affordability

Rising costs in electricity, water and related services disproportionately burden low-income and rural households. UK data show that poorer families spend a far larger fraction of their budget on essentials. For example, the lowest-income decile spent 12.8% of total expenditure on food (non-alcoholic) in 2023/24, versus just 8.7% for the highest decile [65†L462-L471] . Even starker is energy: bottom-decile households spent 10.5% of income on electricity and gas in 2023/24, compared to only 4.6% for the top decile [65†L471-L475] . Water’s share is similarly regressive: typical water bills are ~1.6% of income, but among the poorest third they reach ~3.7% [61†L445-L453] . Rural households often face even higher costs: in Ireland, urban families accounted for 18% of electricity demand versus 10% for rural dwellers [1†L136-L141] , reflecting sparser, less efficient living with higher per-capita bills.

If data centre expansion pushes up utility prices or diverts resources, these low-income and rural groups will suffer most. Increased wholesale power prices from higher demand and infrastructure costs will feed into retail bills. Because energy billing is largely flat-rate for small consumers, any percentage jump hits the budget of a poor family far harder than that of an affluent one. Studies from the US exemplify this: in data-centre-heavy regions (e.g. Northern Virginia), utility bills for average customers have risen far faster than inflation, and low-income households frequently face arrears or even disconnections [41†L219-L227] [41†L240-L249] . Although UK households aren’t on US-style contracts, the same principle holds: a 1% rise in energy prices consumes a larger share of a poorer

household's income. Food and heating trade-offs will become acute for struggling families if electricity prices climb further.

Likewise, water price rises or scarcity induced by new industry demand would stress the poorest. In the UK and Ireland, there is already talk of “water poverty”: around 2 million UK households (especially in Wales and the North) spend over 5% of income on water/sewerage [61†L445-L453] . Additional strain on supplies (e.g. forcing rationing or the need for costly new reservoirs) would ultimately be borne by consumers through higher bills or taxes. Farms would also be at risk: in dry periods, irrigation cuts could threaten rural incomes and food prices. In short, any upward pressure on utilities tends to regressively affect those least able to pay – exactly the demographic least likely to benefit from data centre buildout (which mostly brings short-term construction jobs and tax revenue, not ongoing employment [55†L237-L245]).

Who pays for the infrastructure and subsidies matters greatly. Data centre developments often enjoy tax breaks and government support. In the UK, the government announced *AI Growth Zones* that offer huge electricity discounts and faster grid connections for qualifying data centres [69†L305-L314]

[69†L205-L213] . For example, a 500 MW centre in Scotland could save ~£24/MWh on wholesale power costs [69†L309-L318] (up to ~£80M/year [69†L205-L213]) if it stabilizes the grid. While intended to be self-financing (claimed to “cost nothing” to other billpayers by locating near surplus renewables

[69†L321-L327]), such incentives effectively socialize risk: if forecasts overshoot, consumers cover the grid build with their rates. Data centres have also enjoyed business rates relief and low taxes (Ireland's 6.25% tech tax, US state incentives) in exchange for promised local economic benefits. But in many places those benefits are limited: for instance, Loudoun County (NoVA) realizes ~£700M in DC taxes but 100x that in utility infrastructure costs [55†L222-L231] . Planning regimes may need to claw back costs (e.g. requiring deposits or grid-reinvestment clauses) to avoid ordinary citizens footing the bill for private speculation [37†L219-L227] .

Indeed, policy options must confront equity head-on. Any relief extended to data centres – whether via subsidies, tax breaks or cheap power – ultimately comes from somewhere. Often it reduces revenue for local services or raises bills for others. UK regulators are already proposing reforms to ensure data centre operators take on full market risk: tougher credit tests, grid connection charges, or building their own lines [37†L221-L227] . Similar principles apply to water and land: data centres should ideally pay for water infrastructure upgrades or help fund community resources if they draw heavily on local supply. Otherwise, nearby residents face higher water rates or droughts.

Case Studies

- **Ireland:** The “poster child” for data centre impacts. Ireland's data centers (many in Dublin area) consumed ~21% of national electricity in 2023 [11†L47-L51] – more than households or transport – and this is projected to rise to ~30% by the mid-2030s [12†L285-L293] . After 2021 the Irish regulator effectively paused new DC grid connections until operators could prove they would not overload the grid (e.g. by adding on-site generation) [12†L260-L263] [35†L112-L119] . The government has also signaled that DC expansion will not jeopardize decarbonisation or housing; an interim policy links any new connections to carbon-free or dispatchable backup [11†L47-L51] [35†L112-L119] . Nevertheless, concerns persist: urban (esp. Dublin) grid nodes are near capacity [35†L112-L119] , and rural parts of Ireland may face higher grid charges to compensate. Water stress is also acute – parts of the country have drought-prone aquifers – so additional industrial draws are controversial. Urban Irish households now spend roughly 18% of metered electricity (vs 10% for rural) [1†L136-L141] , underscoring how even within Ireland, impacts vary; further rate increases would affect lower-income and off-grid homes hardest.

- **Netherlands:** After years of rapid DC growth (mostly around Amsterdam), the Dutch government imposed a moratorium on any new “massive” data centres (over 70 MW) until a national plan is ready [50†L79-L87] [50†L102-L103] . Amsterdam itself now bans all new data centre construction beyond prior commitments [53†L55-L64] , citing scarce space on the electricity network and in the city. ING’s analysis warns that the Dutch economy could “lose billions” if expansion halts [57†L57-L64] , even though data centres currently account for only ~3.3% of the Netherlands’ power and 0.08% of water [57†L80-L83] . This highlights a trade-off: technical constraints (and local housing/land pressures) versus the economic prize of being a digital hub. The Netherlands is addressing this with stricter spatial planning (e.g. confining new centres to wind-proximate sites [50†L94-L100]) and promoting reuse of waste heat. Without such measures, energy and water price rises – funded by consumers – could stifle other industries or raise living costs nationwide.
- **Germany:** German data centres have grown around Frankfurt (Europe’s #3 hub) and Berlin, but plans now bump against grid limits. AlgorithmWatch reports that new high-power connections in Frankfurt may not be available until the mid-2030s, with network operators adopting proportional allocation for >3.5 MW projects [52†L148-L157] [52†L158-L163] . Berlin too switched to competitive allocation in 2024 [52†L168-L176] . Utilities like Mainova are racing to double Frankfurt’s transmission capacity in the 2030s [52†L175-L184] , but by then many DC projects may have chosen on-site solutions. In fact, data centres in Frankfurt have begun building their own gas-fired generators (bypassing grid bottlenecks) [52†L190-L199] . Local activists warn this brings noise and pollution: Northern Virginia’s thousands of diesel backups create asthma risks [55†L262-L271] , and similar issues have arisen in Europe’s DC clusters. Germany is debating policies to curb such externalities – for instance, limiting diesel generator usage. Planning barriers are rising: municipalities have seen citizen pushback on new sites (citing climate and resource use [52†L168-L176]), and federal discussions are underway on tougher building standards.
- **United States (Virginia & Texas):** The US offers a cautionary tale. Northern Virginia (“Data Center Alley”) hosts roughly 300 DCs, with dozens more proposed [55†L214-L223] , driven by state tax incentives. The local grid has strained under the load. Dominion Energy (Virginia) projects data centre peak demand to reach 13.4 GW by 2038 (more than twice 2020 levels) [41†L255-L260] . Utilities in Virginia and Texas have begun recovering enormous grid costs via rate increases: one analysis shows Virginia’s prices grew ~267% from 2019–2024 [39†L169-L177] (driven largely by infrastructure for DCs), while US utilities sought ~\$29 bn in rate hikes in early 2025 [39†L158-L166] . These costs ultimately fell to consumers, contributing to home energy bills 27% above 2019 levels nationally [39†L169-L177] . In Texas, water use by DCs is now under scrutiny: one study forecasts Texas data centres will use ~49 billion gallons in 2025 and ~399 billion gallons by 2030 – enough to draw Lake Mead down 16 feet in one year [55†L277-L285] . Many Texas centres have already switched to recycled water or reuse schemes, but urban and agricultural users worry about future cuts. The Virginia case also highlights social impacts: Loudoun County collects ~€900 M/year in DC taxes [55†L223-L231] , yet data centres employ only a few hundred ongoing staff, meaning most infrastructure costs (grid, water, roads) are amortized by local taxpayers and consumers.

Policy Options and Equity Recommendations

Addressing the affordability dimension requires proactive policies. First, **cost internalization** is key: data centre operators should bear the true system costs they impose. This means requiring developers to fully fund necessary grid and water upgrades (via deposits, capacity auctions or “build-and-transfer”

infrastructure programs [69†L225-L234]), rather than socializing them. The UK's emerging *AI Growth Zone* model moves in this direction by letting companies build their own transmission lines or by prioritizing credible projects [69†L225-L234] [37†L221-L227] . Regulators should also tighten merit criteria: requiring staged build-outs, de-listing speculative projects, and clawing back subsidies if promised load fails to materialize. In Ireland, for example, new connection offers now mandate on-site generation or storage to offset the DC's grid draw [35†L112-L119] . Similar mechanisms in the UK/EU could reduce stranded-asset risk: e.g. forfeitable deposits for undeveloped projects, or liability for idle grid capacity.

Second, **progressive utility support** should protect vulnerable users. Any residual rate increases from data centres should be cushioned by targeted relief. This could take the form of enhanced social tariffs or rebates for low-income households (mirroring programs in France/Spain) to offset higher electricity or water rates. Alternatively, revenue from special DC levies or licence fees could be ring-fenced into energy rebate funds. The principle is clear: those least able to pay (bottom deciles, rural poor) should not shoulder the burden of an industrial policy. For example, a small per-kWh surcharge on DC use (charged only to those sites) could finance energy bill assistance for the poorest 20% of households. Such measures would mimic successful “lifeline” rates for essential utilities elsewhere, ensuring that growth in one sector does not tip disadvantaged families into fuel or water poverty.

Third, **environmental safeguards** must be enforced. Any new data centre expansion should be contingent on strict efficiency and sustainability. This means mandating the use of advanced cooling (e.g. closed-loop or renewable-sourced cooling) that minimizes freshwater use [23†L162-L167] , and requiring DCs to operate only on 100% carbon-neutral power (direct renewables or accredited offsets). The UN has urged tech firms to commit to 100% renewable power by 2030 [45†L1584-L1593] ; governments could make this a legal precondition for planning permission or grid connection. Requiring waste heat recovery is another equity option: DCs should be obliged to supply adjacent residential or industrial heat networks wherever feasible, sharing their thermal “free energy” with the community. Dutch ING analysts specifically recommend linking DC approval to sustainable energy usage (e.g. co-locating near offshore wind with heat reuse) [57†L80-L88] . Finally, regulators should institute **transparent monitoring**: operators ought to report their water and energy use publicly, so communities can gauge the real impact.

Fourth, **land-use and planning coordination** can preserve fairness. Local governments should balance DC development against other priorities (housing, agriculture, nature). The Amsterdam and German cases illustrate choosing other community needs over endless data growth [53†L55-L64] [52†L168-L176] . Planning authorities might limit siting in high-demand zones (restricting new DCs where the grid or water supply is weakest) and instead guide them to surplus areas (e.g. brownfield sites near renewables). This spatial strategy – akin to the UK's Growth Zones – can help capture benefits (like using wind power) and avoid overtaxing one region.

In summary, equitable policies should ensure “**who pays**” is clearly defined and progressive. If data centres truly lower system costs (by absorbing off-peak renewables, as Growth Zone theory suggests), those savings should be passed on transparently. If not, the industry should face full market rates and taxes just like any large factory. Some revenue neutrality rules (as in the UK's proposed AI tariffs) aim to avoid loading costs onto consumers [69†L321-L327] , but such guarantees are fragile if political conditions change. It is better to build robust social protections: protecting vulnerable consumers now through billing reforms, and investing DC-generated tax revenue back into communities. Only by weaving together grid planning, fiscal policy and social welfare can governments make the data centre boom compatible with affordability for all.

In conclusion, the data centre expansion across the UK, Ireland and Europe through 2035 is real and significant. Yet its impacts will be uneven. Households in the lowest income deciles – especially in rural or drought-prone areas – will bear the brunt of any higher bills or resource constraints. Policymakers must therefore couple any data centre growth with strong equity safeguards: internalize all infrastructure costs, target subsidies to those in need, and enforce sustainability to protect future generations. Otherwise, the hidden price of “free” data storage will be shouldered disproportionately by those least able to afford it.

Sources: Official energy and utility data, peer-reviewed analyses, regulator reports and expert studies were used throughout. Key figures: UK National Grid/Ofgem (2025–26), EirGrid/CER (2024–25), S&P Global/451 Research (2025), IEA and US think-tank reports (2023–26). Specific citations are given inline.
